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RESEARCH ARTICLE

Online Rapid Job Analysis and Evaluation Using Particle Swarm Optimized Random Forest

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ABSTRACT Manual laborers are often at an increased risk for work-related musculoskeletal disorders (WMSDs) due to improper work postures or the lifting of excessively heavy objects. Therefore, conducting an effective ergonomic assessment of workers is crucial for enhancing productivity and minimizing the occurrence of WMSDs. The Ergonomic Posture Risk Assessment (EPRA) is a widely used method for this assessment. However, traditional EPRA methods rely on human or sensor input, leading to subjective bias, instability, and reduced accuracy. This study addresses these issues by proposing an objective machine-learning approach. It employs a non-intrusive computer vision technique for posture capture, enabling rapid analysis of the worker's activities through Random Forest analysis. The dataset for risk assessment is generated from the worker's skeletal joint posture data, collected using Movenet Thunder in conjunction with an inertial motion capture device and the Rapid Entire Body Assessment (REBA) standard. The Particle Swarm Optimization Random Forest (PSO-RF) model is then utilized to predict risk scores for various postures, incorporating limb length ratios to tackle challenges associated with observing torsional joints. The model's effectiveness in detecting poor posture is subsequently evaluated. The findings indicate that the PSO-RF model successfully identifies poor postures and computes the corresponding REBA scores with 89% accuracy, 93% precision, 91% F1 score, 89% recall, and a kappa value of 82%. This research demonstrates that the machine learning approach, utilizing computer vision and Random Forest, can effectively conduct EPRA to prevent musculoskeletal injuries, providing a data-driven and accurate method for enhancing workplace safety and health.

INDEX TERMS WMSDs, ergonomics, pose recognition, posture angle solving.

I. INTRODUCTION

Work-related musculoskeletal disorders (WMSDs) associated with occupational activities have emerged as a prevalent concern in the field of occupational health [1], [2]. According to the World Health Organization's (WHO) analysis of the global burden of disease data, there has been a significant rise in the number of individuals affected by WMSDs, with approximately 1.71 billion people worldwide experiencing these conditions, impacting diverse age groups [3]. WMSDs profoundly impact workers' quality of life and work efficiency while causing irreversible damage to anatomical structures including muscles, tendons, ligaments, cartilage, joints, and nerves [4], [5]. According to another report, WMSDs persist as the primary occupational ailment in the European Union, impacting workers across diverse industries and roles to a growing degree [6]. Hence, timely recognition of WMSDs risks among employees and the adoption of preventive actions are of paramount importance [7].

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A. PROBLEM STATEMENT

Ergonomic Posture Risk Assessment (EPRA) serves as a vital tool in predicting the impact of WMSDs to preempt potential risks like extreme postures and repetitive movements, facilitating the implementation of preventive measures [8]. Traditionally, researchers assess risk levels by scrutinizing and documenting work postures, loads, and environmental factors using methods such as Rapid Upper Limb Assessment (RULA) [9], Rapid Entire Body Assessment (REBA) [10], and the Occupational Work Analysis System (OWAS) [11], chosen based on task specifics. However, these observational methods harbor subjectivity, potentially compromising assessment stability. Discrepancies in scores assigned by observers for the same posture pose a risk of underestimating or overestimating WMSD predictions [12]. While wearable sensor devices have been explored for objective EPRA, their invasiveness and cost hinder widespread adoption [13], [14].

B. RESEARCH OBJECTIVE

In light of these challenges, this study introduces a novel EPRA method leveraging Particle Swarm Optimization Random Forest (PSO-RF). The objective is to harness machine learning (ML) techniques grounded in the REBA standard to devise stable and highly accurate rules for assessing WMSDs risks. This research makes significant contributions by: (1) capturing operators in work videos with motion capture devices to establish a dataset for holistic posture evaluation, (2) enhancing model detection by integrating limb length ratios with the REBA methodology, (3) optimizing RF parameters using PSO to uphold assessment precision, (4) dynamically monitoring operator states to furnish assessment scores, and (5) comprehensively refining the performance metrics of PSO-RF. Additionally, this study juxtaposes the proposed model against prior research methods and ML techniques, showcasing the innovation and efficacy of this novel approach.

The paper is structured into seven sections. The introduction sets the stage for the study, with related work detailed in the second section. Section three delves into the machine learning aspects, while section four outlines the research methodology. The fifth section unveils the experimental design and data processing techniques, with section six showcasing the experimental results and ensuing discussions. Finally, the seventh section encapsulates a summary and outlines future prospects.

II. RELEVANT THEORIES

The conventional EPRA method entails ergonomic engineers observing and documenting factors like work postures, loads, and environmental conditions following assessment guidelines [12]. Tools such as surveys or checklists are then employed to assess and grade risk levels [15], [16]. For example, the authors in [17] utilized checklists to evaluate work postures. The authors in [18] administered standardized questionnaires through interviews with 8,000 Italian workers. The authors in [19] integrated subjective questionnaires with ergonomic assessments to assess the risk of WMSDs among employees at a TFT-LCD manufacturing company in Taiwan. These methods enable immediate scoring of work postures and offer recommendations for modifications [18], [20]. Additionally, this approach can be time-intensive. Consequently, researchers have devised methods employing contact sensors as a substitute for manual assessments.

The contact sensor method relies on motion capture devices, which gather essential data by positioning sensors at joint locations on the human body. For instance, the authors in [13] and [21] developed a wearable IMU sensor system and applied a Convolutional Long Short-Term Memory (LSTM) model to assess the risk of WMSDs among construction workers. The authors in [22] used surface electromyography and neural networks to predict hand posture and grip strength. The authors in [14] proposed an innovative solution that integrates pressure insoles with trunk IMUs to enhance the assessment, monitoring, and prevention of lower back injury risks across diverse industries. This method produces objective results with high accuracy and is often used as a benchmark for validation. However, its long-term suitability is constrained by sensor detachment and interference with workers' routine tasks. As a result, researchers are now turning their attention to non-contact approaches.

Camera-based non-contact methods allow for posture assessment without disrupting the normal tasks of workers. For instance, the authors in [23] utilized the K2RULA software for EPRA, which is a semi-automatic RULA assessment software based on the Microsoft Kinect V2 depth camera. This software offers real-time identification of poor postures and offline analysis capabilities. Depth cameras pose challenges due to their cost, algorithm complexity, and hardware demands. In contrast, regular cameras are cost-effective and user-friendly, rendering them a more promising avenue for advancement. Detection techniques utilizing regular cameras leverage cutting-edge computer vision progress [24]. These methods commonly employ pose estimation models like OpenPose [25], [26], PoseNet [27], and MoveNet [28], [29] for EPRA [30]. Through pose estimation, researchers extract varied body data, such as joint angles, from operator videos or images. Subsequently, assessment guidelines or machine learning algorithms are utilized to evaluate posture risks. For instance, Lin et al. utilized image-based motion capture techniques to acquire 2D joint angle information using OpenPose. They developed a decision tree system that automatically selects the appropriate assessment scale and calculates risk scores [16]. The authors in [26], [31], and [32] also used OpenPose to obtain 2D keypoints for evaluation. The authors in [33] applied machine learning techniques based on the REBA standard for ergonomic assessment and proposed a Decision Tree Analysis Scheme for Ergonomics Assessment of Work Postures (DTAS-EAWP). The authors in [24] in their study comparing performance and accuracy differences between OpenPose-based methods and Microsoft Kinect-based methods, suggested that utilizing OpenPose for estimating body joint angles to assess RULA and REBA scores holds significant potential for further development. The examples provided highlight the promising potential of utilizing computer vision technology for EPRA.

III. MECHINE LEARNING

Risk prediction encounters challenges like vast data volumes, intricate spatiotemporal relationships, and category similarities [34], [35]. Machine learning methods excel in adaptability, real-time performance, generalization, and robustness, adjusting dynamically through ample training data for risk prognostication, facilitating real-time or near-real-time monitoring. Presently, popular algorithms including Support Vector Machine (SVM), k-Nearest Neighbors (KNN), Decision Trees (DT), and Random Forest (RF) are extensively employed in risk prediction and classification [16], [33], [36]. For instance, The authors in [37] utilized SVM to discern various postures from virtual images and identify them in real-world settings.

A. RANDOM FOREST

RF algorithm, an ensemble learning technique, enhances predictive capability by combining the forecasts of multiple DT. RF is notable for its strong resistance to overfitting, its ability to handle high-dimensional data, and its capacity to assess feature significance. By employing a bootstrap sampling method for constructing each tree, RF generates training datasets for individual decision trees through random sampling with replacement from the overall dataset. The samples that are not selected during this process are referred to as out-of-bag (OOB) samples. Randomness is a defining characteristic of the Random Forest, not only in the sampling of training data but also in the selection of features during the construction of decision trees. Instead of considering all features for split nodes, a random subset is chosen to determine the optimal split, promoting diversity among decision trees and reducing the risk of overfitting. By averaging the predictions from all decision trees, as shown in equation (1), the Random Forest produces a consolidated outcome that reflects the collective predictive insights, ensuring high accuracy. The process of forming a Random Forest is illustrated in Figure 1.

$$f\ (x) = \frac{1}{k}\sum\nolimits_{i=1}^{k} g(x) \tag{1}$$

where $f(x)$ is the average value of the tree, k is the number of trees, and $g(x)$ is the output of each tree.

B. PARTICLE SWARM

PSO stands out as a stochastic search algorithm leveraging population cooperation, commonly applied in machine learning algorithms for optimization tasks. Noted for its swift convergence, minimal parameter dependencies, and straightforwardness [38]. PSO excels at thoroughly exploring search spaces to uncover global optimal solutions, making it ideal for hyperparameter optimization in RF. By defining an objective optimization function and iteratively improving fitness

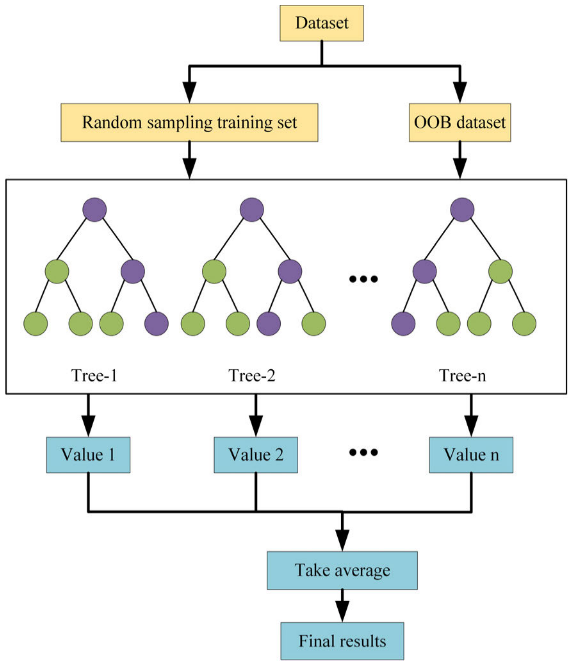

FIGURE 1. The architecture of the random forest.

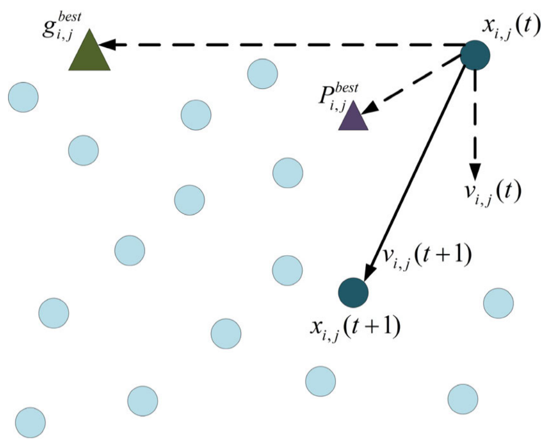

FIGURE 2. The architecture of the particle swarm algorithm.

values, the PSO algorithm refines performance. Through particle interaction and information exchange, the algorithm converges towards each particle's optimal position.

According to the principle of particle behavior, a PSO model is established, and the framework diagram of PSO is shown in Fig. 2. Firstly, several massless particles N are initialized in the parameter value space D, and each particle i has only two attributes: velocity v and position x. Then, each particle searches for the optimal solution individually in D dimensional space, sets the fitness function $f(x)$ as the optimization objective, and $f(x)$ selects the accuracy rate as the fitness function. After t iterations, the fitness function value of each particle is calculated, and the local optimal position is updated if the current fitness function value is better than the historical optimal value $p_{i,d}^{best}$, while the global optimal position is updated if the current fitness function

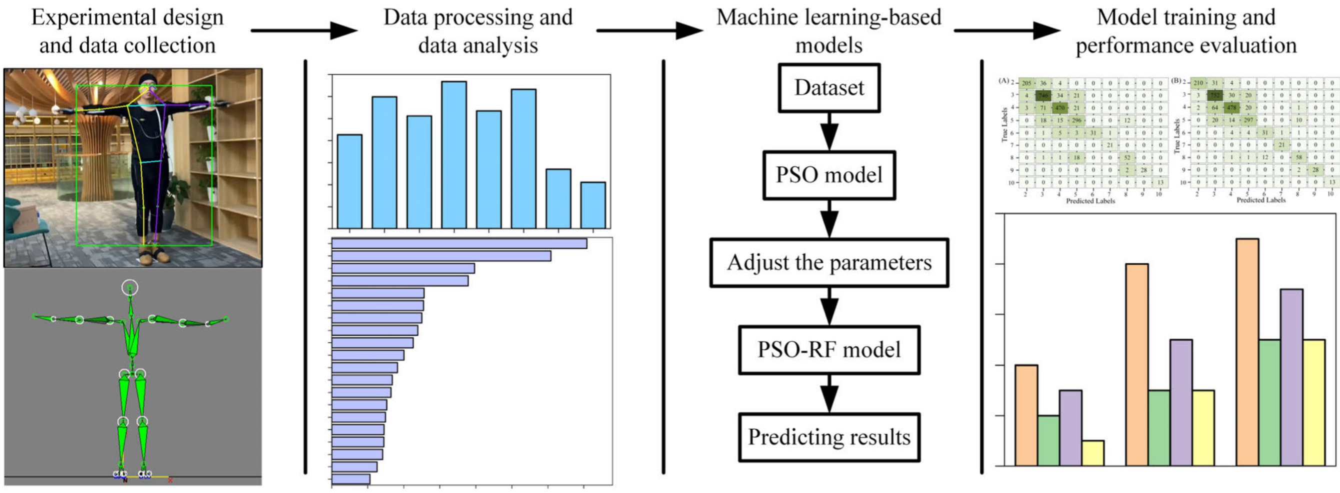

FIGURE 3. Research architecture.

value is better than the global historical optimal value $g_{i,d}^{best}$, and the global optimal position is updated. The solution at the optimal fitness experienced by all particles in the whole particle swarm is taken as the global best position $g_{i,d}^{best}$. According to Fig. 2, it can be observed that the moving direction of the particle in the next iteration is affected by the current velocity $v_{i,d}(t)$, local optimal position $p_{i,d}^{best}$ and global optimal position $g_{i,d}^{best}$, and the velocity update formula of the particle is as follows:

$$v_{i,d}(t+1) = \omega v_{i,d}(t) + c_1 r_1[p_{i,d}^{best} - x_{i,d}(t)] + c_2 r_2[g_{i,d}^{best} - x_{i,d}(t)] \tag{2}$$

where c_1 and c_2 are acceleration coefficients (also called learning factors), generally taken as $c_1 = c_2 = 2$. $d = 1, 2, \ldots, D$ denotes the current dimension. r_1 and r_2 are random numbers sampled uniformly distributed over two [0, 1] intervals to ensure that the algorithm does not lose diversity in the iterative computation. The inertia weighting factor, denoted ω is utilized to regulate the influence of previous speeds on the current speed within a range of 0.4 to 0.9. A higher weight amplifies the global search capability of the algorithm, thereby accelerating the search speed; conversely, a lower weight prioritizes local search capability and consequently leads to a slower search speed. Typically, an initial emphasis is placed on global search, followed by a subsequent focus on local search in each iteration cycle. To achieve this transition smoothly, the inertia weight coefficient is commonly linearly decreased using formula (3):

$$\omega = \omega_{\max} - (\omega_{\max} - \omega_{\min}) \frac{t}{T} \tag{3}$$

where $\omega_{\max}$ and $\omega_{\min}$ are the upper and lower boundaries ω, and T is the maximum number of iterations. The position update formula of the particle is as follows:

$$x_{i,d}(t+1) = x_{i,d}(t) + v_{i,d}(t+1) \tag{4}$$

After updating the particle positions, the current fitness $f(x)$ is computed for each particle and the local optimal position $p_{i,d}^{best}$ and the global optimal position $g_{i,d}^{best}$ are updated:

$$p_{i,d}^{best} = \begin{cases} p_{i,d}^{best}, f\left(x_{i,d}(t+1)\right) \leq f\left(p_{i,d}^{best}\right) \\ x_{i,d}(t), f\left(x_{i,d}(t+1)\right) \geq f\left(p_{i,d}^{best}\right) \end{cases} \tag{5}$$

$$g_{i,d}^{best} = \arg\max_{x} \left\{ f\left(g_{i,d}^{best}\right), f\left(p_{1,d}^{best}\right) \ldots, f\left(p_{N,d}^{best}\right) \right\} \tag{6}$$

The optimization termination is generally based on two criteria: (1) the algorithm reaches the maximum number of iterations T, and (2) the algorithm achieves an acceptable satisfactory solution, indicated by a fitness value difference between the optimal solutions of consecutive iterations that fall below a certain threshold.

IV. RESEARCH METHODS

Figure 3 illustrates the framework of the proposed method, which includes experimental design and data collection, data processing and analysis, a machine learning-based model, model training, and result evaluation. This section clarifies the specific implementation of the machine learning model, detailing the formulas relevant to model training and performance assessment. Following this, the next section explores the aspects of experimental design and data collection, including volunteer selection, experimental equipment, and settings, while also elaborating on the methodologies for data processing and analysis. Additional detailed information is provided below.

A. PARTICLE SWARM OPTIMIZATION RANDOM FOREST MDEL

This study employs the Random Forest algorithm as the primary classification method. The model's hyperparameters. specifically, n_estimators and max_depth—are randomly generated prior to model training. However, variations in parameter configurations during the prediction phase can

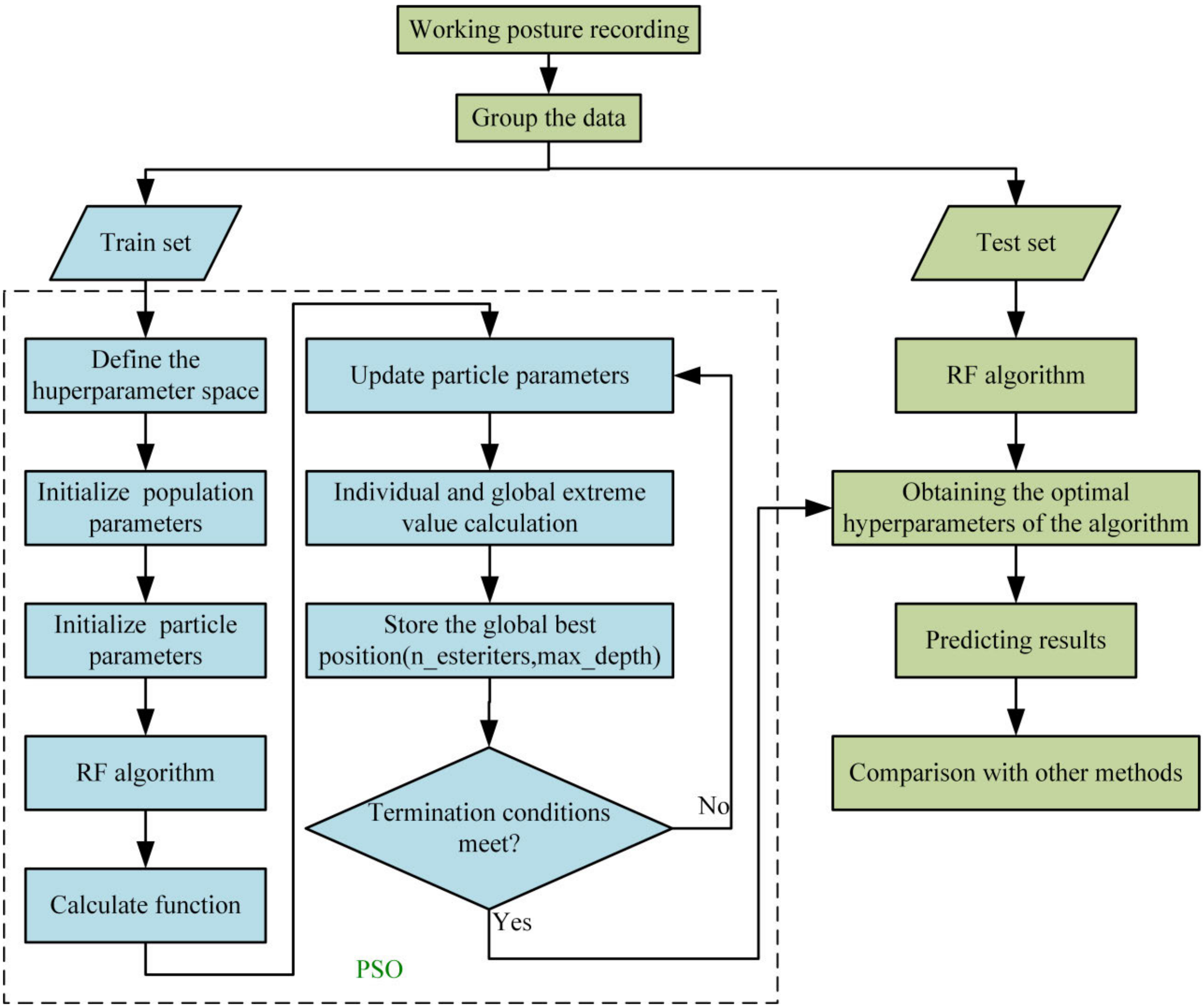

FIGURE 4. Flowchart of the model of PSO-RF.

significantly impact prediction accuracy. Insufficient values for max_depth or n_estimators may compromise the model's classification performance, while excessively large values can extend runtime and reduce classification efficiency. To address this issue, PSO is utilized to optimize the n_estimators and max_depth hyperparameters within the Random Forest model. The PSO-RF algorithm is introduced for classifying human body postures and halts optimization upon reaching a predefined number of iterations. Detailed parameter settings for the algorithm are provided in Table 1, and the flowchart is illustrated in Figure 4. Here, random numbers are generated within a specified interval.

B. EVALUATION METRICS

The classifier's performance evaluation involved multiple metrics, such as Precision, Recall, F1-score, and Accuracy. Additionally, Mean Absolute Error (MAE) was used to measure the variance between estimated joint angles from the pose estimation model and those recorded by the motion capture device. The Kappa coefficient was employed to gauge the agreement between model predictions and actual values. Equations (7)-(12) outline the mathematical expressions for these metrics.

TABLE 1. Model parameter settings.

Variable	Value	Trends of change
c_1, c_2	2	Fixed
r_1, r_2	[0,1]	Random
ω	[0.5,0.9]	Degression
N_estimators	[10,250]	Random
Max_depth	[20,40]	Random

$$\Pr ecision = \frac{\delta}{\delta + \beta} \tag{7}$$

$$\text{Re}call = \frac{\delta}{\delta + \alpha} \tag{8}$$

$$F - score = \frac{2\delta}{2\delta + \beta + \alpha} \tag{9}$$

TABLE 2. Apparatus dimensions.

	Box	Chest	Stool	Rack
Long	20	30	50	60
Wide	20	9	40	45
Tall	20	9	40	230

$$Accuracy = \frac{\delta + \gamma}{\delta + \gamma + \beta + \alpha} \tag{10}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|y_i^* - y_i\right| \tag{11}$$

$$\kappa = \frac{Acc - p_\varepsilon}{1 - p_\varepsilon} \tag{12}$$

where $\delta, \gamma, \beta, \alpha$ are computed from the confusion matrix, true positive, true negative, false positive, and false negative, respectively, N is the number of test samples, y_i^* and y_i are the predicted and target values, respectively, p_ε is the probability of agreement computed based on the confusion matrix;

$$p_\varepsilon = \frac{1}{N^2}\sum\nolimits_k n_{kp}n_{kt} \tag{13}$$

where n_{kp} and n_{kt} are the number of predicted samples in the k row and the number of target samples in the k column.

V. EXPERIMENTAL DESIGN AND DATA PROCESSING

A. EXPERIMENT DESIGN

This study focused on the primary manual handling tasks performed by workers in the printing industry. Daily activities involving the manipulation of printing materials were observed and summarized. Laboratory experiments were conducted, taking into account influential factors such as ambient temperature and lighting conditions that could impact the experimental outcomes. All experiments were completed within a single day. The experiment utilized a six-tier storage rack, a stool, and a movable box and chest, with the equipment dimensions detailed in Table 2. The experimental setup is illustrated in Figure 5.

During the operation, the operator wore motion capture equipment, while the recording device was securely mounted on a smartphone stand positioned 2.2 meters away from the stool. The stool was aligned with the storage rack, which was located 3 meters away, and its height was set at 1.6 meters.

The experiment involved five male Chinese students aged between 24 and 26, all of whom provided informed consent, had no history of chronic musculoskeletal disorders or surgeries, and demonstrated proficiency in the assigned tasks. Real-life operational challenges, such as the obstructed frontal view caused by the storage rack, were taken into account during the recording process. Consequently, actions were recorded from the operator's side view. The action sequence consisted of two task cycles, as illustrated in Figure 6. In each task cycle, the operator retrieved a box and a crate from six different heights on the rack and placed them on a chair positioned two meters away. Subsequently, both items were returned to their original positions on the rack.

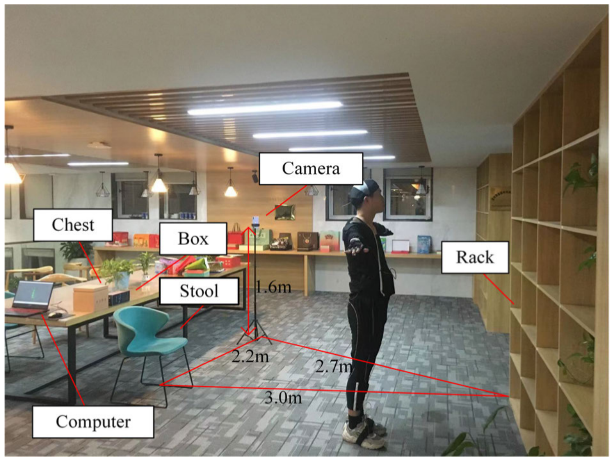

FIGURE 5. Experiment layout.

The box was lifted using bilateral hand manipulation, while one-handed stick manipulation was employed for the crate. Each task cycle lasted approximately 30 seconds, resulting in a total video duration of around 60 seconds covering both cycles. To ensure data variety, five videos were recorded. The sampling frequency was set at 30 frames per second to accurately capture motion details. Synchronization between the recording device and the motion capture equipment was achieved through an external Bluetooth connection.

B. DATA PROCESSING

The task postures were captured using an RGB camera (iPhone 11) and an inertial motion capture device (3DSuit full-body Motion Capture System). This system employs 17 inertial 3D orientation sensors to accurately track the movements of each major body segment, generating BVH (Biovision Hierarchy) files. Renowned for its precision and resistance to interference, this system is widely recognized in research circles. Movenet, developed by IncludeHealth, is a Google-based inference model available in two versions: Lightning, optimized for performance, and Thunder for accuracy [39]. In this study, the Movenet Thunder model was utilized to accurately detect the 2D coordinates of body joints. The angles between these joints were calculated using the law of cosines. Furthermore, the inertial motion capture system was employed to determine the precise positions of the sensors, allowing for the calculation of accurate 3D coordinates for each joint. To ensure robust training labels in the dataset, a combination of the law of cosines and the REBA standard was applied. Detailed insights into this methodology will be presented in the following sections.

1) PROCESSING OF MOTION CAPTURE DATA

The BVH files from the motion capture system contain bone names and rotational coordinates for limb joints. Using the BVH converter tool, three-dimensional coordinate data for each frame's skeleton was extracted, enabling the computation of joint flexion and torsion angles, as detailed in Table 3.

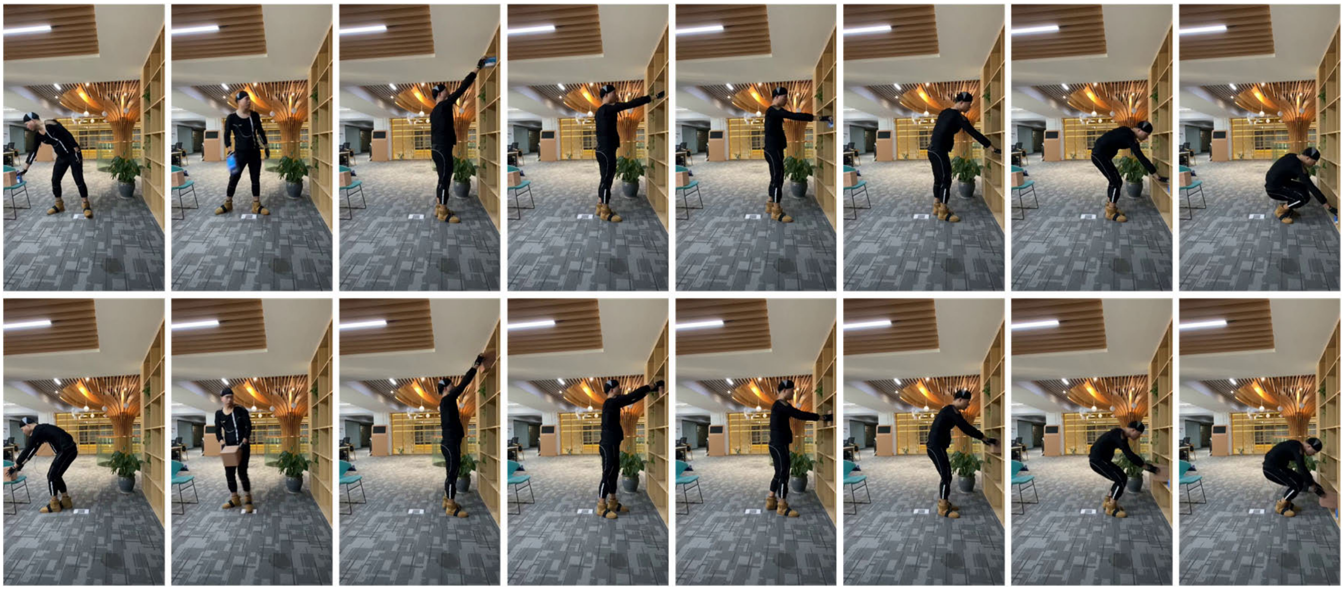

FIGURE 6. Operational task.

Trunk torsion was calculated as the angle between segments 4-5 and 8-9, projected onto the ground, as illustrated in Figure 8. These angles were then translated into assessment scores based on the REBA [10] standard, as illustrated in Figure 7. The experiments were conducted on a flat surface, with the weights of the items involved being under one kilogram, resulting in a load score of 0. The scoring system ranges from 1 to 15, with higher scores indicating an increased risk of WMSDs. Scores between 1 and 3 suggest a low risk, 4 to 7 indicate a moderate risk, 8 to 10 signify a high risk, and 11 to 15 represent very high risk levels. Once the score exceeds 8, it is crucial to promptly adjust the task posture.

2) PROCESSING OF VISUAL DATA

Firstly, the two-dimensional coordinate data of the key points captured by Movenet Thunder (see Figure 9) undergoes interpolation. To prevent erroneous data from impacting machine learning training, unrecognized key point data in the video frames is removed. Subsequently, a linear spline interpolation method is employed to fill in the missing values. Linear interpolation is performed between each pair of adjacent data points using linear functions, effectively connecting these linear functions, the interpolated curve ensures continuity and maintains a consistent slope at every data point. For a given set of n + 1 data points $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$, linear interpolation between every two neighboring points (x_i, y_i), (x_{i+1}, y_{i+1}) is performed using the linear function in (14):

$$f_i\ (x) = \frac{y_{i+1} - y_i}{x_{i+1} - x_i}(x - x_i) + y_i \tag{14}$$

To ensure both value and slope continuity in the interpolation function at each internal node, a series of linear functions is developed to represent the interpolation between adjacent data points, thereby forming the complete interpolation function.

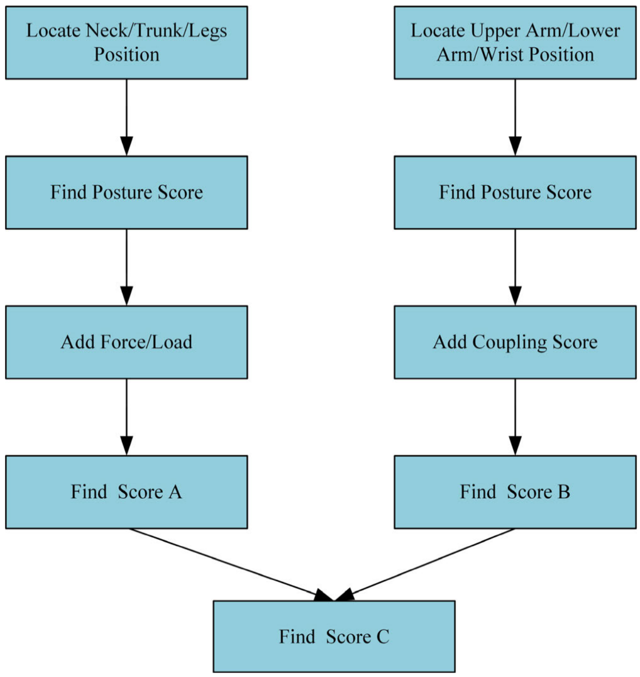

FIGURE 7. REBA standards.

Subsequently, the joint flexion angles are computed for the interpolated data following the methodology outlined in Table 3. However, due to the complexities of projecting three-dimensional images onto a two-dimensional plane and accurately measuring rotation angles, an alternative method is employed to calculate joint torsion angles. It has been observed that limb length influences trunk and arm rotation. By taking into account participants' body shapes and their proximity to the recording device, this research establishes the maximum limb length (MLL) by identifying the limb

TABLE 3. REBA joint angles for motion capture device skeleton data.

Angle name	abridge	Joint involved
Left elbow flexion	LEF	∠2,3,4
Right elbow flexion	REF	∠5,6,7
left shoulder flexion	LSF	∠3,4,8
Right shoulder flexion	RSF	∠6,5,9
Left shoulder abduction	LSO	∠3,4,5
Right shoulder abduction	RSO	∠4,5,6
Left Knee Flexion	LKF	∠8,10,12
right Knee Flexion	RKF	∠9,11,13
trunk flexion	TF	∠mid(4,5),mid(8,9),mid(10,11)
trunk twisting	TT	∠45，89 projector
Neck Flexion	NF	∠1，mid(4,5),mid(8,9)
left wrist flexion	LWF	∠14,2,3
right wrist flexion	RWF	∠6,7,15
Joint numbering according to figure 6		

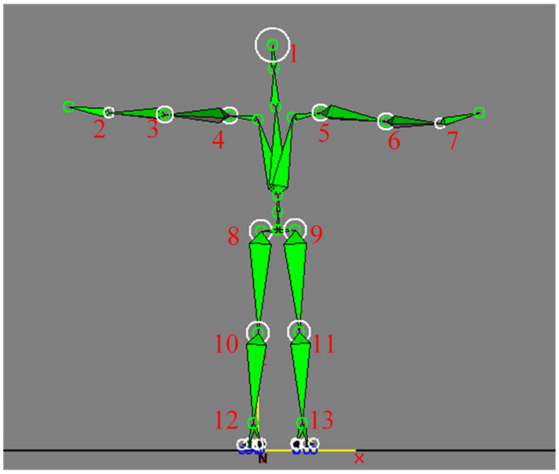

FIGURE 8. Mobilized skeleton data.

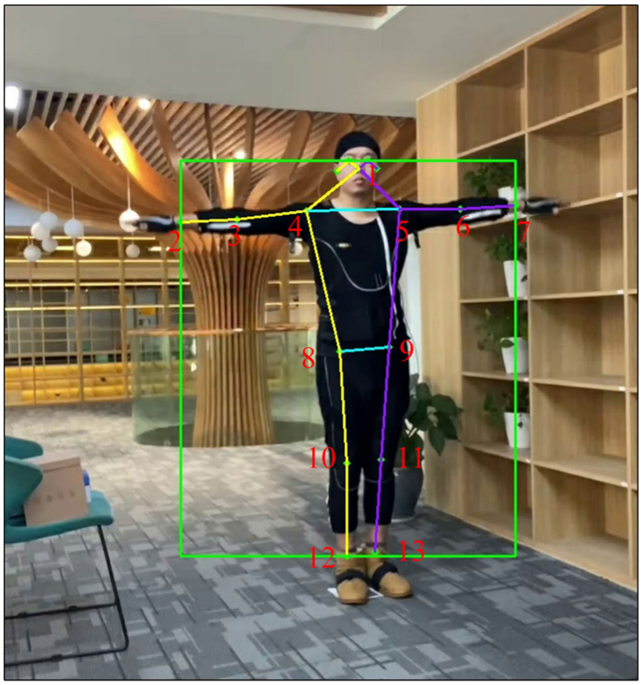

FIGURE 9. Movenet thunder skeleton data.

length when participants face directly toward the recording device (see Figure 9). The limb length ratio (LLR) for each frame is then calculated using Equation (15), which represents the ratio of the limb length (LLPF) to the MLL. Twelve limb length ratios are documented for various body segments. These ratios, in conjunction with the joint flexion angles, determine the joint angles for each video frame. Due to the challenges of accurately deriving wrist angles from photographs, a score of 1 is assigned based on video observations.

$$LLR = \frac{MLL}{LLPF} \tag{15}$$

The dataset comprises eight joint flexion angles and twelve limb length ratios calculated using Movenet Thunder. The dataset labels correspond to REBA scores derived from motion capture data. A comparative analysis was conducted between the joint flexion angles obtained from the motion capture system and those acquired through Movenet Thunder.

In Figure 10, MAE of the differences in joint angles is illustrated. Since a single RGB camera was used for recording, all captured joint angles pertain exclusively to flexion angles.

The average MAE across all joint angles is 9.6°. Notably, the left shoulder and left elbow joints exhibit smaller errors compared to their right counterparts, possibly due to partial occlusion of the operator's right side was recorded exclusively from the left side. The knee joint angles exhibit a minimal MAE of less than 6°, with negligible errors in the data for both knee joint angles. In contrast, the MAE for the right shoulder and right elbow joint angles is relatively larger, yet remains below 14°. These results indicate that the MAE for upper limb joints exceeds that of lower limb joints and the neck joint; however, the overall MAE values remain within an acceptable range.

3) DATA DESCRIPTION

Table 4 illustrates the distribution of the collected data, which comprises 10,663 frames, with all evaluation scores falling within the range of [2, 10]. The data is divided into a training

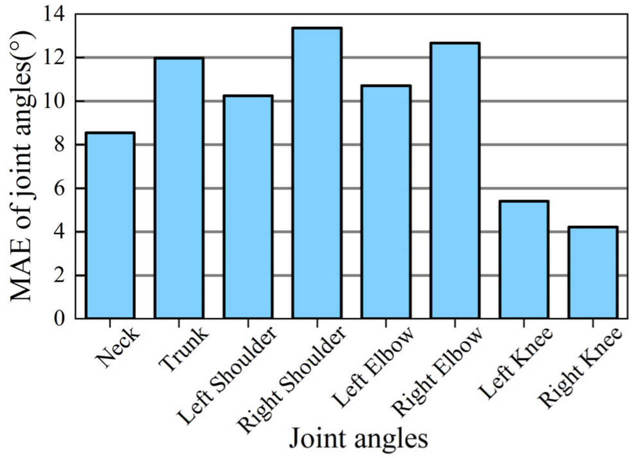

FIGURE 10. MAE of joint angles.

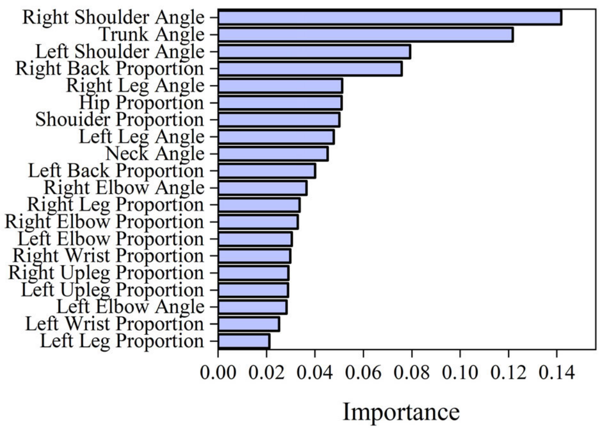

FIGURE 11. Attribute importance analysis.

TABLE 4. Data distribution.

Assessment score	Data volume	Train set data	Test set data
2	1029	784	245
3	3905	3100	805
4	2945	2380	565
5	1733	1392	341
6	199	158	41
7	118	97	21
8	350	278	72
9	126	96	30
10	78	65	13
Total	10663		

TABLE 5. Classification properties of PSO-RF.

	Precision	Recall	F1-score
2	98%	86%	91%
3	87%	93%	90%
4	90%	85%	87%
5	84%	87%	86%
6	100%	76%	86%
7	95%	100%	98%
8	81%	81%	81%
9	100%	93%	97%
10	100%	100%	100%
Accuracy			89%

set and a test set in a 4:1 ratio. Predominantly, the scores are concentrated in the [2, 5] range, with an evaluation score of 3 being the most frequent, accounting for approximately 37% of the total. Conversely, instances with a score of 10 are the least common, occurring in only 78 cases.

As part of the interpretability analysis of RF model, this study examined the feature importance of each input factor. Figure 11 demonstrates that the joint angles of the right shoulder, trunk, left shoulder, right knee, and limb length ratio of the right trunk significantly influenced the outcome. Notably, the angles of the shoulder and trunk joints significantly contributed to the model's predictions, with importance values of 0.12 and 0.14, respectively. Additionally, four limb length ratios ranked among the top ten in importance, underscoring their critical role in representing torsion joint angles in this study. In contrast, leg joint angles had a minimal impact on the final prediction score. This observation suggests that variations in evaluation scores primarily stem from changes in upper body angles, which aligns with established principles of ergonomic assessment.

VI. RESULTS AND DISCUSS

A. EXPERIMENTAL RESULTS

The optimization of the RF model using PSO is significantly influenced by the number of particles and iterations employed in the process. This study found that varying the iteration count from 20 to 60 produced nearly equivalent results; therefore, the research opted for 20 iterations. The particle count was systematically adjusted in increments of 10 within the range of 10 to 100. The analysis indicated that an optimal configuration was achieved with 80 particles, along with n_estimators set to 176 and max_depth set to 29. Figure 12 illustrates the confusion matrices before and after the RF optimization on the test dataset. The vertical axis denotes the actual classes, while the horizontal axis represents the predicted classes. Cells located along the main diagonal indicate correctly classified instances, whereas off-diagonal cells reflect misclassifications. Notably, misclassifications were particularly prevalent for instances with a score of 4. Following the optimization process, the number of misclassifications decreased from 95 to 86. Importantly, both classifiers successfully identified instances with scores of 7 and 10, despite their lower frequency. The classification performance of the PSO-RF model is detailed in Table 5, which presents a performance range from 76% to 100%. The lowest performance value was recorded in the Recall metric for a score of 6. The average metrics for precision, recall, F1-score, and accuracy were reported as 93%, 89%, 91%, and 89%, respectively.

Table 6 presents the classification performance of RF model prior to optimization. The overall classification performance ranges from 72% to 100%, which is slightly inferior to that of PSO-RF model. Notably, the metrics associated

TABLE 6. Classification performance of RF.

	Precision	Recall	F1-score
2	97%	84%	90%
3	85%	93%	89%
4	89%	83%	86%
5	82%	87%	85%
6	100%	76%	86%
7	95%	100%	98%
8	79%	72%	75%
9	100%	93%	97%
10	100%	100%	100%
Accuracy			87%

TABLE 7. Classification performance of PRF.

	Precision	Recall	F1-score
2	94%	83%	88%
3	82%	92%	87%
4	88%	79%	83%
5	81%	84%	83%
6	100%	78%	88%
7	91%	100%	95%
8	82%	65%	73%
9	100%	93%	97%
10	76%	100%	87%
Accuracy			85%

with a score of 8 demonstrate relatively inadequate outcomes. The average values for precision, recall, F1-score, and accuracy are reported as 92%, 87%, 89%, and 87%, respectively. A comparative analysis of the two models indicates that PSO significantly enhances the overall accuracy of the RF model. Furthermore, regardless of the sample size, there is a marked improvement in precision, recall, and F1 score across all classification categories.

The classification performance of the RF model (PRF) is presented in Table 7, without incorporating limb length ratios. To address the errors resulting from the absence of torsion angles, this research employed the method outlined in Table 3 to process the two-dimensional data. As shown in Table 8, the model exhibits classification performance ranging from 65% to 100%, with its lowest performance significantly lagging behind that of the PSO-RF model at 81%. The precision and recall values range from 76% to 100% and from 73% to 100%, respectively. However, for scores of 4, 5, and 8, none reached the threshold of 90% or higher. The average precision, recall, F1 score, and accuracy values are 88%, 86%, 87%, and 85%, respectively.

In contrast to the data presented in Table 5 and 6, there has been a noticeable decrease in all performance metrics of the model. Figure 13 provides a detailed overview of the comparative assessment of the four performance metrics across the three classifiers. Both PSO and limb length ratios play a crucial role in enhancing the model's precision, recall, F1 score, and overall accuracy.

B. DISCUSSION

This research presents an innovative system that integrates computer vision techniques using Movenet with motion capture systems to conduct PSO-RF analysis by the REBA standard. A distinctive feature of this system is its incorporation of limb length ratios to predict the likelihood of workers developing WMSDs during occupational tasks. Using the manual handling activities of printing workers as a case study, the system analyzed a total of 10,663-time frames to assess the associated risk levels. The majority of these time frames were found to fall within REBA scores of [2] and [3], with an analysis of attribute importance highlighting the critical influence of shoulder positioning on REBA scores and the subsequent risk of WMSDs. Deprivation experiments were conducted to validate the effectiveness of the proposed algorithm, resulting in approximately 4% improvements in precision, recall, F1 score, and accuracy. Thus, this study provides a new perspective on limb length ratios and introduces a novel PSO-RF risk analysis methodology, serving as a significant resource for identifying potential risk patterns associated with WMSDs.

TABLE 8. Summary of EPRA covered in this study.

Study	Estimation method	Performance metrics
(1)Ding, et al[40].	Attention CNN	Kappa=73%
(2)Rodrigues, et al[41].	3D-AJA	F1-score=75.9%
(3)Lee, et al[42].	Deep Learning	Kappa=82%
(4)Zhao, et al[43].	CLN	F1-score=70.8% Accuracy=81.2%
(5)This study	PSO-RF	Kappa=85% Precision=93% Recall=89% F1-score=91% Accuracy=89%

A comparative analysis of recent studies on WMSDs alongside the proposed methodology, which includes procedural steps and performance metrics, is summarized in Table 8. It is evident that using a single camera for pose estimation and Kinect data collection, as referenced in the source (2), introduces inaccuracies during the conversion from 2D to 3D coordinates, adversely affecting the overall experimental accuracy. In contrast, the current study employs a streamlined and efficient 2D pose estimation approach that circumvents the complexities associated with conversion and inference processes. The methodologies discussed in sources (1), (3), and (4) all involve processing and extracting information from 2D pose data. The pose detection and analysis system proposed herein demonstrates significant advancements across multiple dimensions compared to contemporary models. Specifically, the performance metrics indicate that for kappa: me > (3) > (1), F1-score: me > (2) > (4), and accuracy: me > (4). Notably, only a limited number of studies provide a comprehensive evaluation of all performance metrics, whereas this study exhibits outstanding performance across all assessed metrics.

This study further investigates alternative machine learning methodologies for the classification of EPRA. A comparative analysis was conducted on SVM, KNN, DT, and Multi-Layer Perceptron (MLP) to evaluate their respective performances. The results of this comparative analysis,

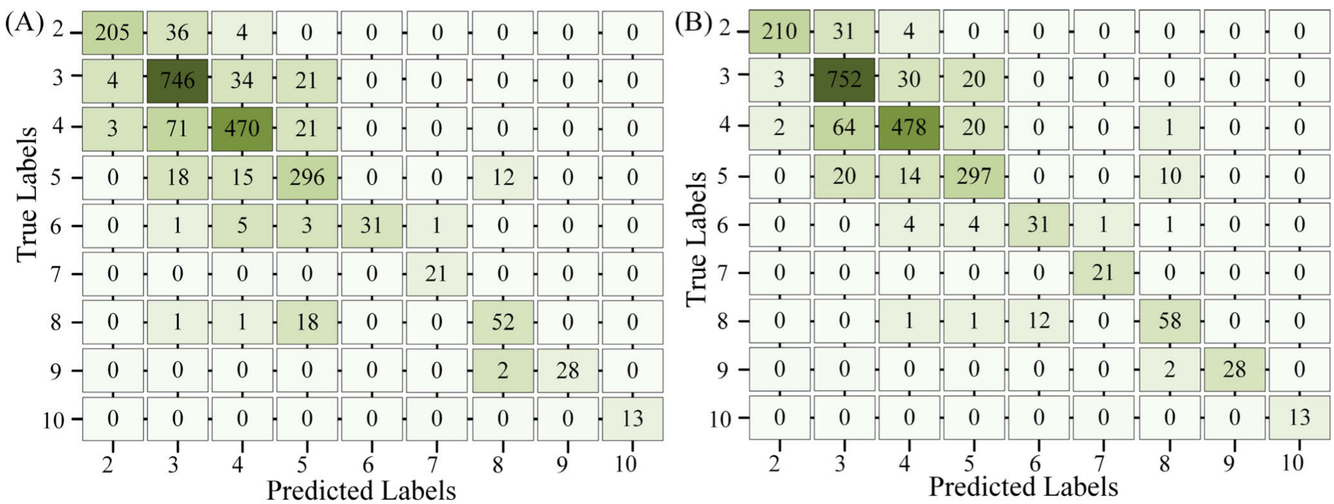

FIGURE 12. Comparison before and after optimization (A) RF confusion matrix (B) PSO-RF confusion matrix.

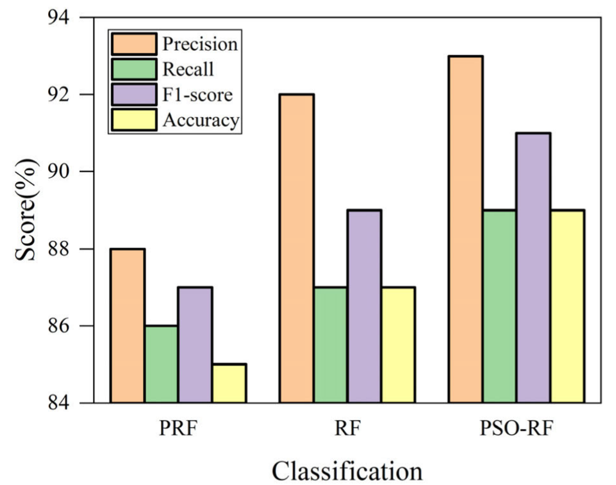

FIGURE 13. Comparison of RF before and after optimization.

TABLE 9. SVM, DT, KNN, MLP classification performance.

	Precision	Recall	F1-score	Accuracy
SVM	38%	40%	39%	73%
DT	80%	80%	80%	83%
KNN	83%	83%	83%	82%
MLP	78%	66%	71%	77%

including precision, recall, F1-score, and accuracy for each classification algorithm, are presented in Table 9. Among the evaluated methods, SVM exhibited the lowest performance, with precision, recall, and F1-score ranging from 38% to 40%. In contrast, MLP demonstrated moderate performance across all metrics, with values between 66% and 78%. Both DT and KNN displayed comparable performance levels, approximately 82%. Nonetheless, a significant disparity in performance metrics is observed between these algorithms and the proposed PSO-RF algorithm.

A comparative analysis of traditional methodologies and various machine learning techniques reveals that the approach proposed in this study is distinguished by its objectivity, reliability, precision, and comprehensiveness in evaluation. The high-performance PSO-RF model offers actionable intervention strategies and serves as a valuable reference for addressing WMSDs. By employing machine learning techniques that are devoid of subjective biases in ergonomic assessments and risk predictions, as evidenced by previous research, the automatic quantification of ergonomic risks becomes feasible. This study positions the PSO-RF as an effective tool for the accurate analysis of ergonomic data. Through an analysis of attribute importance, it identifies critical joint angles and introduces limb length ratios to mitigate the limitations associated with single-camera systems, thereby enhancing the accuracy of the evaluation methods. In conclusion, the PSO-RF model effectively characterizes the risks faced by operators in developing WMSDs. Consequently, ergonomic specialists, employees, and employers can collaboratively explore a range of preventive strategies, including modifications to workplace environments, task designs, and enhanced protective measures for vulnerable body parts, to safeguard worker health against WMSDs. The findings of this study provide substantial evidence supporting the implementation of practical interventions and strategies aimed at mitigating WMSDs.

VII. CONCLUSION

This research presents a posture classification methodology that integrates computer vision and machine learning algorithms to automate the assessment of ergonomic postures. By systematically categorizing workers' movements, this approach aims to prevent harmful postures that could jeopardize worker health, thereby reducing the incidence of WMSDs. Evaluations conducted on five male participants with diverse physical characteristics tested the method under various conditions, demonstrating that its posture classification accuracy is comparable to that of motion capture systems. A significant advancement of this study is the application of limb length ratios to address challenges associated with observing twisted joints, thereby enhancing performance through the PSO-RF algorithm. This methodology shows promise in addressing the wide range of human postures and may assist practitioners with limited ergonomic knowledge by facilitating ergonomic risk assessments based solely on video recordings of workers. In summary, ergonomic assessments grounded in computer vision and machine learning enable rapid task evaluations, risk identification, and timely risk mitigation, thereby empowering workers to proactively manage the risk of WMSDs.

Nonetheless, there are opportunities for improvement within this methodology. Data obtained from video recordings are inherently subject to uneven distributions. This study has made partial advancements in the model's ability to recognize minority classes through the application of ensemble learning techniques. Future research will incorporate data augmentation strategies to achieve a more balanced dataset across various samples, potentially utilizing Generative Adversarial Networks (GANs) to generate additional minority class samples for dataset expansion. Furthermore, while the experimental data in this study were collected in a controlled laboratory environment, real-world factory settings present complexities such as diverse backgrounds and factors like camera pixelation that may influence assessment results. Subsequent investigations will involve recordings taken in actual factory environments to broaden the research scope, facilitating the identification of similarities and differences in WMSDs risks associated with various postures. Future initiatives will also apply the PSO-RF method across different types of work in various factories, ultimately leading to the development of a mobile and web-based comprehensive platform for the prevention of WMSDs. This platform will effectively support ergonomic risk assessments aimed at identifying hazardous postures, providing evidence for risk mitigation strategies, and reducing the likelihood of WMSDs development.

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